

A Novel Approach to Image Forgery Detection Techniques in Real World Applications



Dhanishtha Patil, Kajal Patil, and Vaibhav Narawade

1 Introduction

In the past decades, the era of digitalization has made it easier to use a variety of data sources, including images, to conduct scientific experiments, derive valuable insights for business and technology, identify repetitive patterns of data to develop the machine learning models, a popular field in today's world [1]. Among all the varied sources of data available, digital image seems to be one of an emerging category of producing a high impact which will accelerate the research of the particular domain [2].

Significant advancements in computer engineering have culminated in very high resolution capture equipment and substantial breakthroughs in image processing, making picture tampering and forgeries more accessible and straightforward. Some people distort the contents of a picture by modifying parts of it, resulting in image forgery, which is becoming more hard to trace using modern image processing tools [3, 4]. With the aid of advanced computer graphics algorithms and picture editing softwares like Photoscape and Acorn, photographs may now be manipulated without leaving any imprints [5]. To modify picture characteristics such as pixels, size, and resolution, a variety of applications is available [6].

Given the accessibility of powerful media editing, analysis, and production tools, as well as the increasing processing capability of contemporary computers, image altering and generation is now straightforward even for unskilled users. This trend is only likely to continue, resulting in more automated and precise processes becoming available to the general public [1, 7]. Image forgery can be defined as "erroneously and maliciously altering a picture." The concept of image forgery dates back to 1840.

D. Patil (✉) · K. Patil · V. Narawade
Ramrao Adik Institute of Technology, Nerul, Navi Mumbai, India
e-mail: [dhanishthapatil1804@gmail.com](mailto:ghanishthapatil1804@gmail.com)

V. Narawade
e-mail: vaibhav.narawade@rait.ac.in

The earliest altered image was created by Hippolyte Bayard, a French photographer, in which Bayard admitted to attempting himself [8]. The certainty that the picture has not been altered or accessed is known as image integrity. When the integrity of an associated sensitive picture is breached in particularly applications that are critical, such as radioactive and military, a critical procedure might be jeopardised, as a consequence [9].

To counter the issue, digital forensics emerged as a crucial and trending field with a motive of ensuring the detection of digital tampering and manipulation. Because of their capacity to recreate the evidence left by cyber assaults, forensic tools have become a critical tool for Information Assurance [10]. As a result, pictures no longer have the status of being the only definitive record of events. In the lack of an authentication or watermark, digital forensics describes numerous statistical approaches for identifying evidence of digital manipulation [11].

Active and blind or passive image validation methods are available. Digital watermarking and digital signatures, for example, to detect active counterfeiting, include a valid authentication feature in the picture content before delivering it over an unprotected public channel [5]. The active forging approach uses a digital icon and requires pre-processing to create a watermark and embedded or signatures before compiling the picture. Although watermarking is a live intruding disclosure approach in which a certification shape is implanted into the image, most imaging devices lack a watermarking or check module [12]. This data might be added throughout the picture collection process, or afterwards with the help of an appropriate tool. Passive image forgery detection strategies, on the other hand, to detect fabrication, no previous information of the input image is required. Alternatively, these approaches recognize forgeries by looking for changes in the image's fundamental characteristics that may have been introduced during the modification process [13]. Copy-move forgery is copying a piece or region from one position in a photo and transferring it to another site inside the same image in order to conceal or imitate an item or a collection of items and create a fictitious perspective [10]. To hide an existing item in a picture, make a replica of the entity, copy-move forgery is used [2].

Based on the study of pre-existing literature, it was discovered that there are numerous domains in which image manipulation malpractices are carried out, each with a varied immoral, unethical motive. Some of them are briefly explained as follows:

Medical Diagnosis forgery:

Image falsification is a major problem in the healthcare industry. The diagnosis will be incorrect, and the patient will be in life-threatening danger if a visual diagnostic report is manipulated and the attacker employs any kind of forgeries to expand the cancerous region. If a healthcare framework includes an image forgery detection system, it can identify the counterfeit before the diagnostic process begins if medical data is stolen or changed, for example, the patient may experience social humiliation or be let down, while others may gain an unfair advantage. As a result, in a smart healthcare architecture, there should be a system that can verify whether medical data is corrupted during transmission by hackers or intruders [11].

Deepfake:

DeepFakes are artificial intelligence systems that utilise deep learning to replace the likeness of one person in video and other digital media with that of another. Deepfake technology has raised worries that it may be used to produce fake news and deceptive films. The most challenging challenge in the realm of picture forgery detection is detecting fake faces. Phony face pictures may be used to construct fake identities on social networking sites, allowing for the illicit theft of personal information. The false picture generator, for example, may be used to create photos of celebrities with improper material, which might be dangerous.

Social media image manipulation:

In today's digital world, social media platforms play a critical role in news dissemination. They have, however, been disseminating false pictures. On social media platforms like Twitter, forged photos cause deception and negative user sentiments. As a result, identifying fraudulent photos on social media sites has become a pressing requirement [17]. Not every material shared on social media is exactly what it claims to be. People upload and disseminate incorrect information for a variety of reasons. These might include a) personal goals (to acquire recognition or celebrity) or b) intents to sway public opinion or promote specific points of view (marketing, propaganda, etc.) [13]. Image forgery can now be done with a variety of technological tools, computer software, and web applications. As a result, the usage of modified photographs in news portals and, more specifically, social media has proliferated [14].

Criminal investigation evidence forgery:

The reliability of photos is critical in a variety of fields, including forensics, criminal investigations, surveillance systems, and intelligence agencies. Image manipulation is quite prevalent [18]. Manipulating the images is very much possible using the software accessible to the criminals. Because digital photographs contain vital information and are used in various sectors such as authenticating digital photographs is becoming increasingly important in legal issues of investigation and serve as a crucial source of proofs [19]. Image tampering, which is a critical component of a criminal investigation, can lead to legal authorities making defective or improper judgment [15].

Art forgery:

Even for veteran art historians, authenticating paintings may be challenging. It might be difficult to determine the authenticity of an artwork. Forgery detection has traditionally relied on the discriminating talents of "connoisseurs," who can determine credibility based on a prominent art piece, lifestyle, and circumstances [20]. There has been a lot of development in utilizing digital feature extraction techniques to characterize an artist's style. Studies have been able to categorize test paintings as originals or forgeries based on these findings [16].

2 Literature Survey

Forgery detection is a crucial issue in visual forensics, with a large body of literature on the subject and numerous deep learning techniques have dominated in recent years. General-purpose extremely deep architectures produce similar good outcomes in favourable circumstances when given a sufficiently big training set. The goal is to detect patterns that identify transitional zones that are out of the ordinary in comparison to the backdrop in order to locate probable forgeries. This concept is also explored in [21], which uses a hybrid CNN- LSTM architecture to train a binary mask for forgery localisation. However, to train the net, these approaches require comprehensive ground truth maps, which may not be available or exact. Researchers demonstrated and developed numerous methods for verifying picture integrity and detecting various types of image fraud.

Shen et al. [22] proposed a technique for detecting forgeries based on textural characteristics extracted from Grayscale Co-Occurrence Matrices (GCOM) and DCT. The CASIA-V1 dataset had a detection rate of 98%, whereas the CASIA-V2 dataset had a detection rate of 97%. In the quaternion DCT (QDCT) domain, Li et al. [23] proposed a forgery detection method based on Markov. For the CASIA-V1 dataset, the detection rate was 95.217%, while for the CASIA-V2 dataset, it was 92.38%. The CASIA-V1 dataset had a detection accuracy of 97%, whereas the CASIA-V2 dataset had a detection accuracy of 97.5%. The CASIAV1 dataset had a detection accuracy of 99.16%, whereas the CASIA-V2 dataset had a detection accuracy of 97.52% [24]. The CASIA-V1 dataset had a detection accuracy of 98.3%, whereas the CASIA-V2 dataset had a detection accuracy of 99.5% [25]. The Curvelet transform and the LBP were employed by Al Hammidi et al. [26] to identify forgeries, and the findings were 93.4% accurate. For the CoMoFoD dataset, the findings obtained a detection accuracy of 92.22% [27]. He et al. [28] developed a technique for detecting forgeries in both the DCT and DWT domains based on extracting Markov characteristics. For the CASIA-V1 dataset, the findings obtained a detection rate of 93.33%. Kakar et al. [29] proposed a forgery detection technique based on DCT and DWT domain Markov characteristics. For the CASIA-V2 dataset, the findings obtained a 95.5% detection rate. Using a pyramid model and Zernike moments, Ouyang et al. [30] proposed a Copy Move Forgery Detection (CMFD) method. The findings revealed that for scaling ranges of 50 to 200%, the given technique had the best influence on the arbitrary rotation angle. A new block-based CMFD matching method was proposed by Lai et al. [31]. The findings revealed that the proposed technique is more resistant to JPEG picture compression assaults and the addition of Gaussian or salt pepper noises than the traditional algorithm. Saleh et al. [32] proposed a multiscale weber local descriptor-based forgery detection technique. Vaishnavi and Subashini utilised the Random Sampling Consensus method and local symmetry characteristics (RANSAC). For MICC-F600 datasets, the findings reached 90.2% detection accuracy [33]. Rao and Ni used CNN to achieve a detection rate of 98.04% for the CASIA-V1 dataset and 97.83% for the CASIA-V2 dataset. The findings obtained

95% identification accuracy for the MICC-F600 dataset when Agarwal and Verma used the pretrained Visual Geometry Group-net (VGGNet) deep learning model [34].

3 Proposed Methodology

Recent research in computer vision, however, has revealed that CNNs are taking advantage of the rapid rise in the amount of annotated data and significant advances, topics have been made. Convolutional Neural Networks shown in Fig. 1 are scaled up for increased accuracy if more resources are available [35]. The CNN accumulates hierarchical visual characteristics from training dataset, that are subsequently utilised to differentiate between tainted and genuine pictures. Extensive testing has shown that the proposed algorithm outperforms existing technologies [36]. In this paper we have investigated model scaling in depth and attempted to find a model that carefully balances network depth, width, and resolution to improve performance for Image Forgery.

3.1 Dataset

Deep learning-based approaches beat other technologies and solutions for the MICC group dataset, the most well-known and practical datasets for testing copy- move forgery detection methods [10] developed by Amerini et al. MICC data include pictures that have a visible manipulation impact. In MICC F2000 picture's tampered regions are randomly picked regions from the that data itself and level of forgery for particular image in MICC F2000 is around 1.12% of the whole image of 2048×1536 pixels which 700 are tampered and 1300 are original images [36, 37]. For model training and validation, MICC F2000 is utilized, also there are 260 forged picture sets in the CMFD [38]. The CoMoFod and MICC F220 datasets are used for testing. The input images are converted to RGB and scaled to a predetermined size

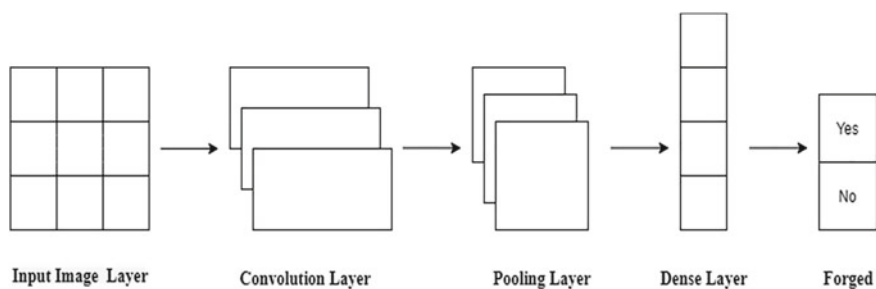


Fig. 1 Basic CNN architecture

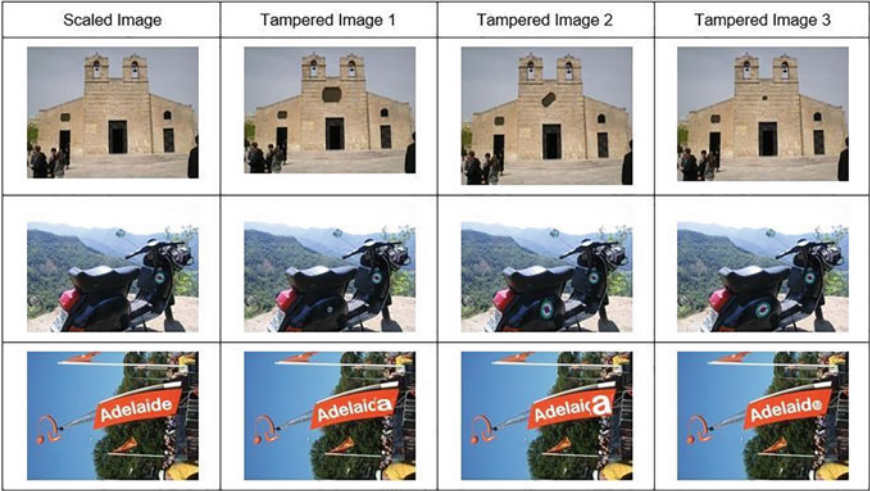


Fig. 2 Images from dataset MICC F2000

without cropping any sections during the data pre-processing step. Figure 2 shows some images from the training dataset that is MICC F2000.

3.2 VGG-19 Architecture

The VGG-19 model is a convolutional model with a wide range of image classification success. Its design consists of three small convolutional filters (3×3), the convolution stride is fixed at 1 pixel, and the padding is similarly fixed at 1. On the supplied dataset, a batch size of 16 is used to train the network. Validation is done with the same batch size. The activation function of the ReLU is implemented. A dropout of 0.5 is used. The RMSprop optimizer is used with a learning rate of $1e-4$. Only vertical variations are allowed by the RMSprop optimizer. When training the model, the Adam optimiser was also utilised to get to the global minima. When training, if you get stuck in local minima, the Adam optimiser will help you get out and attain global minima. For training, we utilised 15 epochs. Validation accuracy of 97.50% is achieved with only 15 epochs.

3.3 VGG-16 Architecture

Instead of a wide range of hyper-parameters, VGG16 concentrated on 3×3 filter convolution layers with a stride 1 and always retained the same padding and maxpool layer of 2×2 filter stride 2. Throughout the design, the convolution and max pool

layers are positioned in the same way. It has two completely linked layers in the end [39]. The network is trained with a batch size of 30 on the provided dataset. ReLU's activation function has been implemented. A dropout of 0.5 is used. The RMSprop optimizer is used with a learning rate of $1e-3$ and a decay factor of 0.75. The Adam optimizer to attain global minima. We used 15 epochs for training. The accuracy of validation is 97.75%.

3.4 *EfficientNet*

EfficientNet is a convolutional neural network design that performs better on ImageNet than existing CNNs in terms of effectiveness and precision, while lowering parameter size and FLOPS by an order of magnitude and scaling approach that uses a compound coefficient to evenly scale all depth, breadth, and resolution parameters [40]. The drop connect rate is set for better regularisation and rebuilt top layers after loading EfficientNetB0. 0.2 dropout rate is chosen. Also the GlobalAveragePooling2D layer is added, since by imposing correspondences between feature maps and categories, convolution structure is created. In addition, the batch normalisation layer is used in the initial training and dropout to set the input unit to 0 at a frequency of at each step throughout the training time, which helps avoid overfitting, and the dense layer with softmax activation is used in the final training and dropout. We used Adam Optimizer to train the model with a learning rate of $1e-3$ and a decay of 0.75. The batch size of 16 and the number of epoch 15 are the same. Validation accuracy of 96% is achieved after 15 epochs. In the second training phase top 20 layers are not froze except the batch normalization layers and further the model is trained on 15 epochs and adam optimizer with learning rate of $1e-4$ which in final obtained accuracy of 98.2%. The training was done on Google Colab with NVIDIA Tesla P100 GPU and 12 GB RAM.

4 Results

4.1 *Evaluation Metrics*

A confusion matrix is a table that describes the results of categorisation problem prediction. Count values are used to sum and break down the number of successful and failed projections by class (Table 1).

True positive (TP): The model's estimated value corresponds to the originality of the image. It can be deduced it is identified that the image as forged.

False negative (FN): The expected output is a false negative, in which the image in input is wrongly labelled negative, despite the fact that the image is unforged.

True negative (TN): The anticipated outcome is a genuine negative when the model

Table 1 General confusion matrix

	Predicted forged(1)	Predicted unforged(0)
Forged(1)	TP	FN
Unforged(0)	FP	TN

Table 2 Confusion matrix for EfficientNetB0

	Predicted forged(1)	Predicted unforged(0)
Forged(1)	249	7
Unforged(0)	0	144

Table 3 Confusion matrix for VGG 16

	Predicted forged(1)	Predicted unforged(0)
Forged(1)	248	8
Unforged(0)	1	143

Table 4 Confusion matrix for VGG 19

	Predicted forged(1)	Predicted unforged(0)
Forged(1)	247	9
Unforged(0)	1	143

predicted value matches to the reality that the picture is unforged. It may be inferred that the machine classified properly.

False positive (FP): Image is incorrectly categorized as an unforged image, despite the fact that it is a forged image (Tables 2, 3 and 4).

From above observations, TNR, TPR is high and FNR, FPR is low. So our model is not in overfit or underfit.

In comparison research, the performance of data classification algorithms is evaluated using metrics such as classification accuracy, sensitivity or recall, precision, and Matthew Correlation Coefficient after the confusion matrix is constructed (MCC) (Tables 5, 6 and Figs. 3, 4, 5).

Table 5 Rate comparison

	TPR%	FPR%	FNR%	TNR%
EfficientNetB0	100.00	4.64	0.00	95.36
VGG 16	99.59	5.30	0.40	94.70
VGG 19	99.59	5.92	0.40	94.07

Table 6 Performance comparison

	Accuracy	Recall	Precision	F1 Score	MCC
EfficientNetB0	0.9825	1.00	0.9727	0.9861	0.9631
VGG 16	0.9775	0.9960	0.9688	0.9822	0.9524
VGG 19	0.9750	0.9960	0.9648	0.9802	0.9473

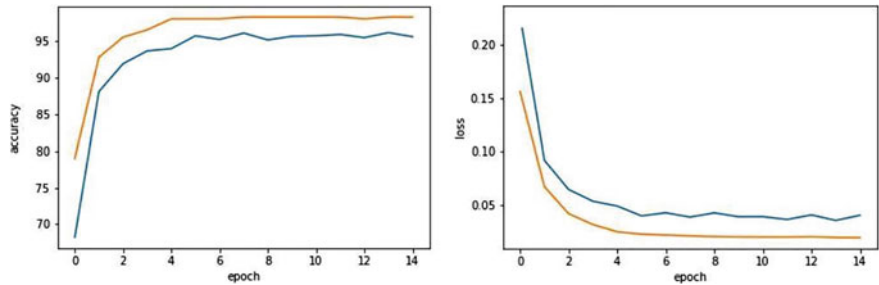


Fig. 3 Performance curves for first phase for EfficientNetB0

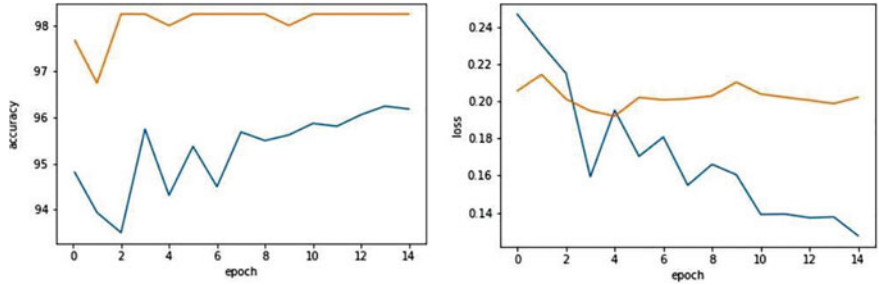


Fig. 4 Performance curves for final phase for EfficientNetB0

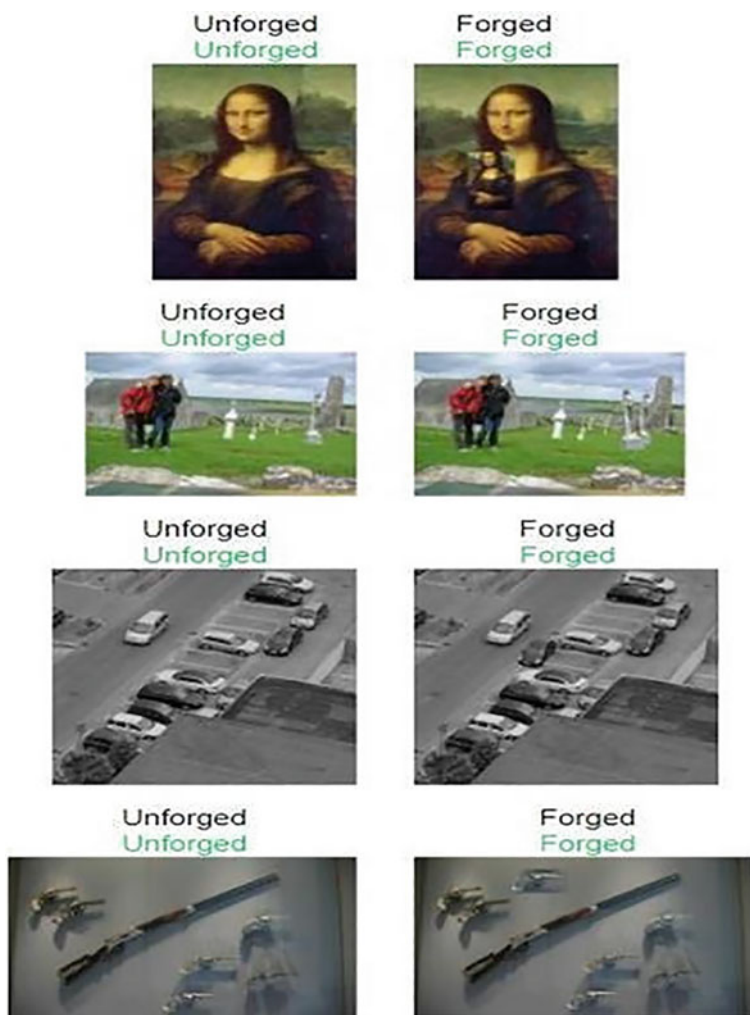


Fig. 5 Final results for most of real time based applications of Image Forgery in fields of Art, Social Media, Criminal Investigation

5 Conclusion

The focus of this research is to show how picture data counterfeiting methodologies may be employed to perform image tampering and falsification. One of the most rapidly developing fields of research is forgery detection utilising passive forgery detection approaches. Various fields being affected by image forgery have also been scrutinized. Analysis of the performance studies on forged images from various realistic domains where image forgery is done to depict the usefulness of this method.

The main objective was to identify the fabrication using copy move technique a variety of CNN architectures and compare them in terms of how accurate their results are. Amongst the Convolutional Neural Network architectures, Efficient-NetB0 achieved the highest validation accuracy of over 98%. The study can further be expanded to develop various methodologies to detect audio forgery and an efficient system can be established.

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